System Analysis & Architectural Design

RCS 305 ? Software Engineering | Riara University Revision Resource

1. Transitioning from Problem to Solution Space

System Analysis investigates ’what’ the business domain requires. Architectural Design determines ’how’ the computing system will fulfill those requirements through structured components, databases, and network interfaces.

2. Structured Analysis Artifacts

? Data Flow Diagrams (DFD): Represent information flow through processes, external entities, data stores, and data flows without implementation assumptions.
? Entity-Relationship Diagrams (ERD): Model persistent relational or document structures, foreign key constraints, and cardinality ratios (1:1, 1:N, M:N).
? State Transition Diagrams (STD): Track reactive system behavior across discrete lifecycle events.

3. Fundamental Design Principles

High Cohesion: A module should focus on a single, well-defined operational responsibility.
Loose Coupling: Modules should minimize direct knowledge and inter-dependencies on other modules, interacting via stable interfaces.

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